

DynamicAllocation-engine

Mathematical Framework and Methodology

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1 Introduction

The DynamicAllocation-engine is a regime-aware multi-asset credit allocation framework. The purpose of the framework is to dynamically allocate capital according to macroeconomic conditions, volatility regimes, and portfolio optimization principles.

2 Analysis

PART I — Regime Identification

2.1 Principal Component Analysis

Suppose

$$X \in \mathbb{R}^{n \times p}$$

denotes the matrix of macroeconomic variables

$$X = f[\text{CPI, Policy Rate, Sovereign Yield}]$$

where

- CPI denotes inflation;
- Policy Rate denotes the central bank interest rate;
- Sovereign Yield denotes the 10-year government yield.

2.2 Standardization

The variables are standardized using Z-scores

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}$$

which yields the standardized matrix

$$Z = \begin{bmatrix} z_{1,\text{CPI}} & z_{1,\text{PR}} & z_{1,\text{SY}} \\ z_{2,\text{CPI}} & z_{2,\text{PR}} & z_{2,\text{SY}} \\ \vdots & \vdots & \vdots \\ z_{n,\text{CPI}} & z_{n,\text{PR}} & z_{n,\text{SY}} \end{bmatrix}.$$

2.3 Covariance Matrix

The covariance matrix is computed by

$$\Sigma = \frac{1}{n-1} Z^T Z.$$

For the present data set, the covariance matrix becomes

$$\Sigma = \begin{bmatrix} 1.0000 & -0.2165 & -0.3320 \\ -0.2165 & 1.0000 & 0.8729 \\ -0.3320 & 0.8729 & 1.0000 \end{bmatrix}.$$

2.4 Eigenvalue Problem

Principal component analysis is obtained by solving

$$\Sigma v = \lambda v.$$

Equivalently,

$$(\Sigma - \lambda I)v = 0,$$

where

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and

$$\det(\Sigma - \lambda I) = 0.$$

The eigenvalues are

$$\lambda_1 = 2.0207,$$

$$\lambda_2 = 0.8604,$$

$$\lambda_3 = 0.1189.$$

Notice that

$$2.0207 + 0.8604 + 0.1189 = 3,$$

which equals the total variance.

2.5 Eigenvectors

The corresponding eigenvectors are

$$v_1 = \begin{bmatrix} 0.356 \\ -0.650 \\ -0.671 \end{bmatrix},$$

$$v_2 = \begin{bmatrix} 0.929 \\ 0.325 \\ 0.178 \end{bmatrix},$$

and

$$v_3 = \begin{bmatrix} -0.102 \\ 0.687 \\ -0.719 \end{bmatrix}.$$

2.6 Principal Components

Therefore, the first principal component is

$$PC_1 = 0.356(\text{CPI}) - 0.650(\text{Policy Rate}) - 0.671(\text{Sovereign Yield}),$$

which explains

$$\frac{\lambda_1}{3} = \frac{2.0207}{3} = 67.36\%$$

of the total variation.

Similarly,

$$PC_2 = 0.929(\text{CPI}) + 0.325(\text{Policy Rate}) + 0.178(\text{Sovereign Yield}),$$

explaining

$$\frac{0.8604}{3} = 28.68\%.$$

Finally,

$$PC_3 = -0.102(\text{CPI}) + 0.687(\text{Policy Rate}) - 0.719(\text{Sovereign Yield}),$$

explaining

$$\frac{0.1189}{3} = 3.96\%.$$

2.7 Conclusion

The first principal component explains approximately 67% of the total variation in the macroeconomic variables and may therefore serve as a regime indicator within the DynamicAllocation-engine. Also, almost all of the information contained in CPI, Policy Rate, and Sovereign Yield can be summarized by just two latent factors.

2.8 K-Means Clustering

Having obtained the principal component scores, observations are grouped using the K-means algorithm. The objective of K-means is to partition the observations into K clusters such that the within-cluster variation is minimized.

Formally, the algorithm solves

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{i \in C_k} \|t_i - \mu_k\|^2,$$

where

- t_i denotes observation i in principal component space;
- C_k denotes cluster k ;
- μ_k is the centroid of cluster k .

The K-means algorithm proceeds iteratively:

1. Initialize K centroids.
2. Assign each observation to the nearest centroid,

$$\arg \min_k \|t_i - \mu_k\|.$$

3. Update each centroid according to

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} t_i.$$

4. Repeat until convergence.

For the present study, the principal component scores were partitioned into three clusters.

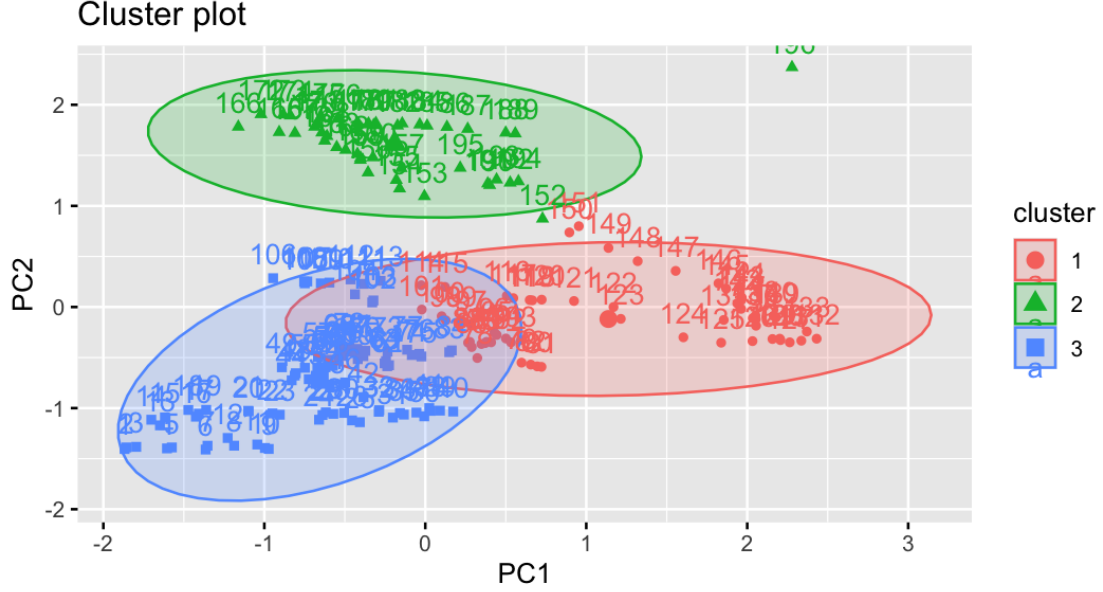


Figure 1: K-means clustering of observations in the principal-component space.

Three distinct clusters emerge from the data. Inspection of the macroeconomic characteristics associated with each cluster suggests three monetary regimes:

1. Eased Monetary Conditions,
2. Tight Monetary Conditions,
3. Mixed Conditions.

Hence, the clustering procedure provides a data-driven classification of macroeconomic environments which will subsequently be used to study ETF performance across regimes.

3 ETF Performance by Regime

Having identified three macroeconomic regimes, the next step is to investigate whether asset performance differs across these regimes.

Suppose ETF j has monthly return

$$r_{t,j} = \frac{P_{t,j} - P_{t-1,j}}{P_{t-1,j}},$$

where $P_{t,j}$ denotes the price of ETF j at time t .

Returns are subsequently grouped according to the previously identified regimes.

3.1 Mean Returns by Regime

For ETF j , the average return within regime k is

$$\bar{r}_k = \frac{1}{n_k} \sum_{i \in C_k} r_i,$$

where

- n_k denotes the number of observations belonging to regime k ,
- C_k denotes the set of observations in regime k ,
- r_i denotes the ETF return.

The empirical results indicate that expected returns are regime-dependent,

$$E[r|R=1] \neq E[r|R=2] \neq E[r|R=3].$$

This observation forms the basis of regime-aware asset allocation.

4 Portfolio Construction

Having established the macroeconomic regimes and documented their effects on asset performance, the next step consists of constructing portfolios dynamically.

4.1 Implied Spread Proxies

Classical credit spreads are unavailable for exchange-traded funds. Consequently, implied spread proxies are defined as excess returns relative to Treasuries (IEF).

For ETF j , returns are computed as

$$r_{j,t} = \frac{P_{j,t}}{P_{j,t-1}} - 1.$$

Implied spread proxies are then defined as

$$Spread_{j,t} = r_{j,t} - r_{IEF,t}.$$

Hence,

$$Spread_{LQD,t} = r_{LQD,t} - r_{IEF,t},$$

$$Spread_{HYG,t} = r_{HYG,t} - r_{IEF,t},$$

$$Spread_{AIGI,t} = r_{AIGI,t} - r_{IEF,t},$$

and similarly for all remaining assets.

4.2 Forecasting Expected Returns

Expected returns are generated using a regime-dependent autoregressive model.

For asset j ,

$$Spread_t = \alpha + \phi Spread_{t-1} + \beta_2 D_{2,t} + \beta_3 D_{3,t} + \varepsilon_t,$$

where

- α denotes the intercept;
- ϕ measures spread persistence;
- D_2 and D_3 are regime indicators;
- β_2 and β_3 capture regime effects.

Given the relatively small sample size and the objective of obtaining a parsimonious forecasting model, an AR(1) specification augmented by regime dummies was adopted.

This specification exploits the empirical persistence of credit spreads,

$$Corr(Spread_t, Spread_{t-1}) > 0.$$

4.3 Forecast Covariance Matrix

Besides expected returns, portfolio optimization requires an estimate of the covariance matrix

$$\Sigma_{t+1}.$$

The sample covariance matrix is computed as

$$\Sigma = Cov(R),$$

where

$$R = [r_{IEF} \ r_{LQD} \ r_{HYG} \ r_{AIGI} \ r_{AHYG} \ r_{BRJP}].$$

Thus,

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{16} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{26} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{61} & \sigma_{62} & \cdots & \sigma_{66} \end{bmatrix}.$$

4.4 Mean-Variance Optimization

At each time period, the optimizer solves

$$\max_w \left(w' \mu - \frac{\gamma}{2} w' \Sigma w \right)$$

subject to

$$\sum_{i=1}^6 w_i = 1,$$

and

$$w_i \geq 0.$$

Here,

- w denotes portfolio weights;
- μ denotes forecast expected returns;
- Σ denotes the covariance matrix;
- γ is the risk-aversion parameter.

4.5 Dynamic Portfolio Allocation

For each month t ,

1. regime probabilities are inferred;
2. expected returns are forecast using AR(1);
3. covariance matrices are estimated;
4. mean-variance optimization is solved;
5. optimal weights are obtained.

This procedure produces a sequence of dynamic portfolio allocations

$$w_1^*, w_2^*, \dots, w_T^*.$$

Average allocations within each regime are subsequently computed to characterize regime-specific portfolio profiles.

5 Performance Evaluation

The performance of the dynamic regime-based strategy was evaluated relative to several benchmark portfolios.

5.1 Jensen's Alpha

The Capital Asset Pricing Model (CAPM) framework was used to assess abnormal performance.

Following Jensen (1968), excess returns are defined as

$$y_t = R_{p,t} - r_{f,t},$$

where $R_{p,t}$ denotes portfolio returns and $r_{f,t}$ the risk-free rate. Similarly, benchmark excess returns are

$$x_t = R_{m,t} - r_{f,t},$$

where $R_{m,t}$ denotes benchmark returns.

The following regression model was estimated,

$$R_{p,t} - r_{f,t} = \alpha + \beta (R_{m,t} - r_{f,t}) + \varepsilon_t,$$

where

- α denotes Jensen's alpha;
- β measures systematic risk exposure;
- ε_t denotes the disturbance term.

5.1.1 Dynamic Portfolio Relative to Static MVO

The CAPM regression yielded

$$\hat{\alpha} = 0.005703,$$

and

$$\hat{\beta} = 1.001283.$$

Hence,

$$R_{Dynamic,t} - r_{f,t} = 0.005703 + 1.001283 (R_{Static,t} - r_{f,t}).$$

The positive and statistically significant intercept ($p = 0.00636$) indicates that the dynamic regime-based portfolio generated approximately 0.57% abnormal return per month beyond that explained by the static mean-variance portfolio.

Furthermore, the estimated beta coefficient is close to unity, suggesting that both portfolios exhibit nearly identical systematic risk exposure.

Consequently, the positive alpha implies that incorporating macroeconomic regimes improves risk-adjusted performance.

5.1.2 Dynamic Portfolio Relative to Equal Weight

Similarly, the dynamic strategy generated

$$\hat{\alpha} = 0.00650,$$

with

$$\hat{\beta} = 0.985.$$

The positive and statistically significant alpha ($p = 0.00403$) indicates that the regime-based strategy outperformed the naïve equal-weight benchmark on a risk-adjusted basis.

Since the estimated beta coefficient remains close to unity, the superior performance cannot simply be attributed to higher systematic market exposure.

5.1.3 Interpretation

Overall, the positive and statistically significant Jensen's alpha estimates indicate that the regime-based dynamic allocation strategy consistently generated abnormal returns relative to both benchmark portfolios.

These findings suggest that incorporating macroeconomic regimes into portfolio construction improves risk-adjusted performance beyond conventional static allocation methods.

A Ordinary Least Squares Estimation

The coefficients are obtained by minimizing

$$\min_{\alpha, \beta} \sum_{t=1}^T (y_t - \alpha - \beta x_t)^2.$$

Taking first-order conditions yields

$$\hat{\beta} = \frac{\sum_{t=1}^T (x_t - \bar{x})(y_t - \bar{y})}{\sum_{t=1}^T (x_t - \bar{x})^2},$$

and

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}.$$